

The transmission line

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Introduction
2. Distributed model
3. Surge impedance loading
4. Lumped transmission line model
5. Line rating

Introduction

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point towards each other from the left and right sides, meeting at a point at the bottom center. A smaller, darker teal triangle is positioned at the very bottom center, partially overlapping the base of the other two.

What will we learn today?

- ▶ The transmission line
- ▶ An introduction to power flow analysis

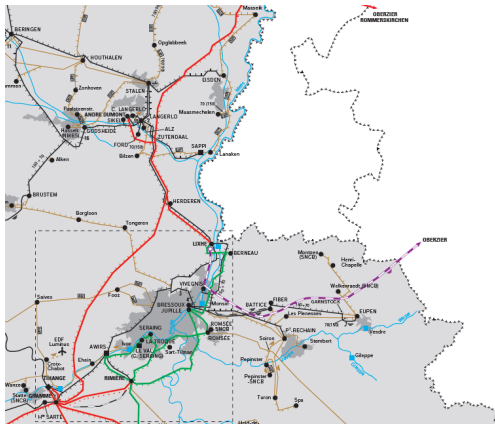
You will be able to do exercises 4.3, 4.4, 4.6, 4.7, 4.8, 4.9, 4.10, 4.11, 4.12, Lab4 (power-flow in python), 5.1, 5.2, 5.5, 5.6 from Ned Mohan's book.

Introduction video link

Definition

- ▶ An (overhead) transmission line is a set of 3 bundles of conductors corresponding to the three phases of the system.
- ▶ Commonly used voltages range from 70 kV to 380 kV in Belgium (more where distances are larger).
- ▶ Minimum distances between conductors depend on the voltage level, and thus electrical properties also depend on the voltage level.
- ▶ Underground cables are more and more used. They can be modeled in a similar way as overhead transmission lines, but they have different properties.

A part of ELIA's network



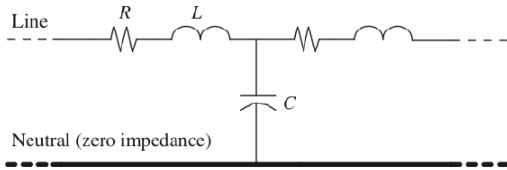
LIGNES AERIENNES		BOVENGRONDSE LIJNEN	
Tension d'exploitation 380kV 220kV 150kV 110kV 70kV		Uitvoeringspanning 380kV 220kV 150kV 110kV 70kV	
Nombre de terres previs 1 1 2 1 2 2 > 2		Aantal draadstellen voorsien 1 1 2 1 2 2 > 2	uitgevoerd 1 1 2 1 2 2 > 2
(voir schéma de référence dans le tableau des composants)		(voir referentiepunten in de samenstellingstabel)	
en construction ou en projet 2 ^{ème} terme en construction ou en projet		in aanbouw of in ontwerp 2 ^{de} draadstel in aanbouw of in ontwerp	
lignes à 2 terres de sensibilité différente tension d'exploitation inférieure à la tension de construction ligne appartenant à un tiers (T)		lijn met 2 draadstellen van verschillende spanningen uitvoeringspanning lager dan de constructiespanning lijn eigendom van een derde (T)	
Tableau des compositions des lignes à plus de 2 terres:		Samenstellingstabel van de lijnen met meer dan 2 draadstellen:	
1 2 x 150 + 1 x 70/110 + 1 x 70	7 2 x 380 (0 x 380)	13 1 x 70/110 + 1 x 70	
2 4 x 150	8 2 x 70	14 1 x 220 + 2 x 70	
3 1 x 150 + 2 x 70/110 + 1 x 70	9 2 x 110 (0 x 110)	15 2 x 150 (0 x 150)	
4 1 x 150 + 2 x 70/110 + 1 x 70	10 1 x 150 + 1 x 70/110 + 1 x 70	16 2 x 150 + 1 x 70/110 + 1 x 70	
5 2 x 150 + 2 x 70/110 + 1 x 70	11 4 x 70	17 4 x 230	
6 2 x 150 (0 x 380 + 2 x 150)	12 2 x 150	18 1 x 380 + 2 x 150 (0 x 380 + 2 x 150)	

CABLES SOUTERRAINS		ONDERGRONDSE KABELS	
380kV 220kV 150kV 110kV 70kV HVDC		380kV 220kV 150kV 110kV 70kV HVDC	
en construction ou en projet câbles parallèles câble appartenant à un tiers (T)		in aanbouw of in ontwerp parallèle kabels kabel eigendom van een derde (T)	

Source: <https://www.elia.be/fr/infrastructure-et-projets/nos-infrastructures>

Transmission line parameters

A *chunk* (a tiny piece) of transmission line can be represented as:



with R , L and C expressed **per unit of length**.

where

- ▶ R represents the series resistance, as small as possible to minimize RI^2 (influence of the frequency and skin effect)
- ▶ the series inductance L models the magnetic coupling between phases
- ▶ the shunt capacitance C models the capacitive coupling between phases
- ▶ a shunt conductance G can be added to model e.g. the leakage current through insulators

Typical cross section of a high voltage overhead conductor

Skin depth (depth at which decay of current density is $1/e$ of surface density):

Material	Frequency [Hz]	depth [mm]
Copper	50	9.4
Copper	60	8.6
Aluminum	50	12.0
Aluminum	60	10.9



Approximate Overhead Transmission Line Parameters

For bundled conductors at 60 Hz.

Nominal Voltage	R (Ω/km)	ωL (Ω/km)	ωC ($\mu\text{F}/\text{km}$)
230 kV	0.06	0.50	3.4
345 kV	0.04	0.38	4.6
500 kV	0.03	0.33	5.3
765 kV	0.01	0.34	5.0



Approximate underground cable parameters

For conductors at 60 Hz.

Nominal Voltage	R (Ω/km)	ωL (Ω/km)	ωC ($\mu\text{F}/\text{km}$)
110 kV	0.05	0.030	95
220 kV	0.04	0.025	115
330 kV	0.03	0.020	130
400 kV	0.02	0.018	150

CU/XLPE/PVC/STA/PVC



CU/XLPE/PVC



CU/XLPE/PVC/SWA/PVC



AL/XLPE/PVC

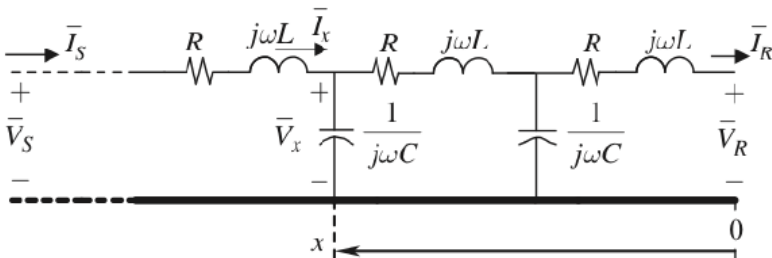


Distributed model

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point upwards from the bottom, meeting at a central point. A smaller, darker teal triangle is positioned at the very bottom center, creating a layered effect.

Distributed parameter representation I

We consider that we are in sinusoidal steady state. On a per-phase basis, the line can be represented as many chunks connected to each other:



Distributed parameter representation II

How do voltage and current evolve as a function of the position on the line?

- ▶ As R is small, let's assume R is considered as *lumped* (a discrete resistive element on one side of the line)
- ▶ $\frac{d\bar{V}(x)}{dx} = j\omega L\bar{I}(x)$
- ▶ $\frac{d\bar{I}(x)}{dx} = j\omega C\bar{V}(x)$

Hence

$$\frac{d^2\bar{V}(x)}{dx^2} + \beta^2\bar{V}(x) = 0$$

with $\beta = \omega\sqrt{LC}$ the *propagation constant*

Solution of the ODE I

The previous equation has a solution of the type

$$\bar{V}(x) = \bar{V}_1 e^{\beta j x} + \bar{V}_2 e^{-\beta j x}.$$

By derivation, the current is

$$\bar{I}(x) = (\bar{V}_1 e^{\beta j x} - \bar{V}_2 e^{-\beta j x})/Z_c.$$

With the **surge impedance**

$$Z_c = \sqrt{\frac{L}{C}}.$$

Solution of the ODE II

The boundary conditions at $x = 0$,

$$\bar{V}(0) = \bar{V}_R = V_R \angle 0,$$

and

$$\bar{I}(0) = \bar{I}_R$$

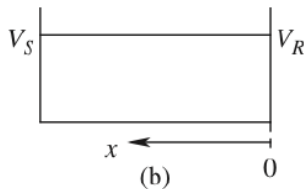
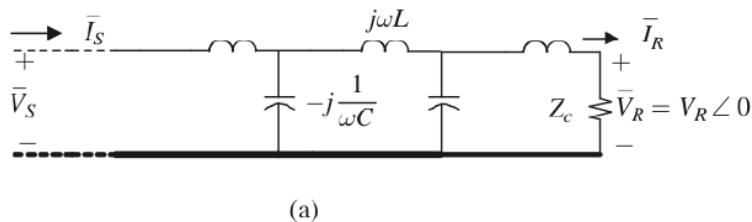
allow to determine constants $\bar{V}_1$ and $\bar{V}_2$, and finally

$$\bar{V}(x) = \bar{V}_R \cos(\beta x) + jZ_c \bar{I}_R \sin(\beta x).$$

Surge impedance loading

Closing the line on the surge impedance Z_c

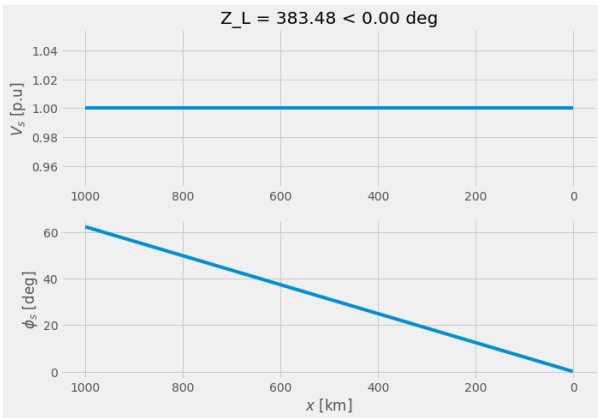
If the line is assumed lossless and we close it with Z_c , assuming $\bar{V}_R = V_R \angle 0$:



then the voltage *magnitude is constant* over the line: $\bar{V}(x) = V_R e^{j\beta x}$, and only the *angle increases with x* . Similar conclusion for $\bar{I}(x)$.

- Why? The reactive power consumed by the line is the same as the reactive power produced, everywhere.

Illustration in Python



SIL, 230 kV line params

See the [Python notebook](#).

Surge impedance loading

Z_c depends on the line characteristics/geometry and is, hence, mainly a function of the voltage level (distances between conductors, etc.).

The surge impedance loading (SIL) is the power drawn by the load Z_c , which depends on the voltage level V_{LL}

$$SIL = \frac{V_{LL}^2}{Z_c}$$

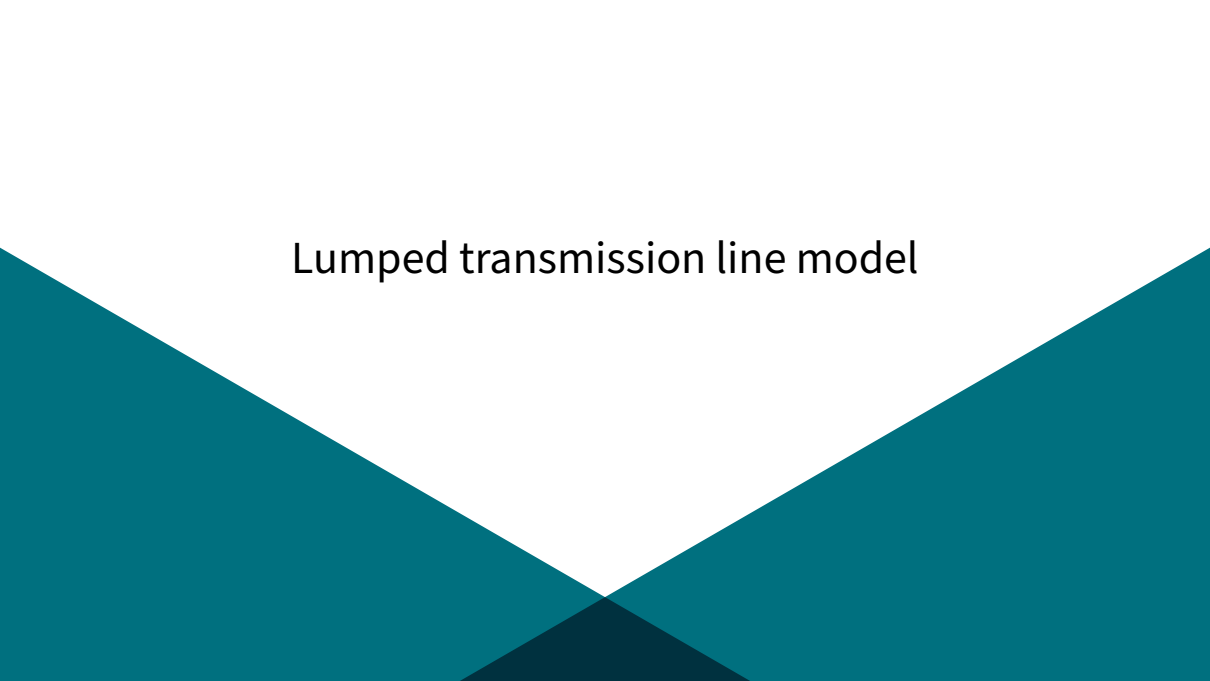
Example: for 500 kV, $SIL \approx 1020$ MW.

Line loadability

The SIL gives an idea of the loadability of a line depending on its length:

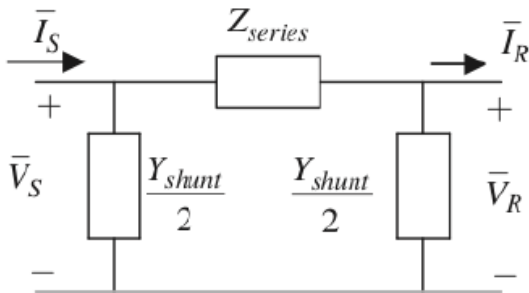
- ▶ short line, $l < 100$ km
 - ▶ load limit = $3 \times SIL$
 - ▶ thermal limit (See Section 5 on Line rating)
- ▶ Medium length line, $100 \text{ km} < l < 300 \text{ km}$
 - ▶ load limit = $1.5 \text{ to } 3 \times SIL$
 - ▶ voltage drop $< 5\%$
- ▶ Long line, $l > 300 \text{ km}$
 - ▶ load limit $\approx 1 \times SIL$
 - ▶ for system stability, the angle difference between line ends should stay $< 40^\circ$, see lecture on Transient stability.

Lumped transmission line model

The background of the slide features a white upper half and a teal lower half. The teal section is composed of two large triangles meeting at a point at the bottom center, with a smaller, darker teal triangle positioned directly beneath that point.

The π model

If l is relatively small (< 300 km), we can *approximate* the line with **lumped** parameters,



with, by manipulation of the previous equations and assuming βl small,

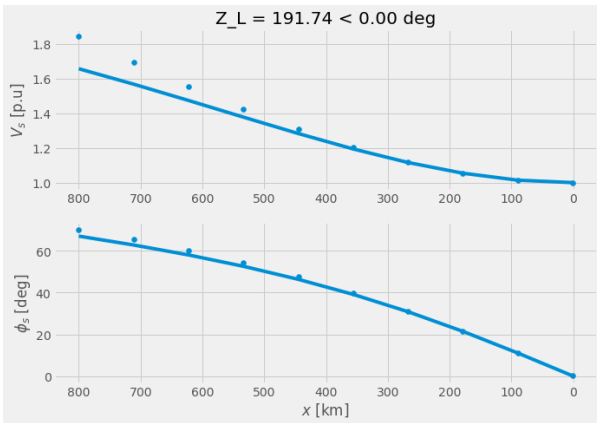
► $Z_{series} = Rl + j\omega Ll$

► $\frac{Y_{shunt}}{2} = j\frac{\omega Cl}{2}$

Remember that R , L and C are per km values.

This π model is symmetrical by design.

Illustration in Python



SIL, 230 kV line params

Dots are obtained with the π model, while solid lines are from the distributed model. The approximation error grows for $l > 300$ km.

See this [Python notebook](#).

Line rating

Static line rating I

The **Static Line Rating** is the maximum continuous current a transmission line can carry under a specific set of predefined, fixed environmental conditions.

This rating is primarily constrained by **thermal limits** to ensure the safe and reliable operation of the line.

► Conductor Thermal Limit:

- The conductor's electrical resistance generates heat (RI^2).
- This heat must be balanced by cooling from the environment.
- An excessive temperature can cause:
 1. **Increased Sag:** Conductor expansion due to heat causes the line to sag, potentially violating minimum clearance requirements to the ground, buildings, or other infrastructure.

Static line rating II

2. **Material Damage:** Prolonged high temperatures can anneal the conductor, reducing its tensile strength and lifespan.

► **Environmental Conditions:**

- The "static" nature of the rating comes from assuming a fixed set of weather parameters.
- **Ambient Air Temperature:** A baseline temperature (e.g., 40°C) is used. Higher temperatures reduce the cooling capability of the air.
- **Wind Speed:** A low, static wind speed (e.g., 2 ft/s at a 45° angle) is assumed to provide minimal convective cooling. Higher wind speeds would allow for more current.
- **Solar Radiation:** A fixed value for solar heat gain is included, assuming a specific level of direct sunlight on the conductor.

Static line rating III

Static line ratings are a conservative and fundamental safety measure. They represent the worst-case continuous scenario to prevent physical damage and clearance violations under predictable conditions.

Dynamic line rating

[Dynamic line rating video link](#)